

Exercise 3:

Fix the provided Pintool in Probe mode called **btranslate.cpp** (located at the course site at: <https://moodle25.technion.ac.il/mod/resource/view.php?id=136832>) so that it manages to successfully translate the provided binary program **cpugcc_r_base.mytest-m64** and generate the correct output file **200.s** using the following pin command:

```
$ <pindir>/pin -t obj-intel64/btranslate.so -- cpugcc_r_base.Oz-m64 200.i -o 200.s
```

Currently, the **btranslate.cpp** pintool fails to properly translate the executable and exits with a “Segmentation fault” error without generating the output file **200.s**.

Use **any** available methods and tools to fix the pintool in order to generate the correct output.

Explain the issue that you found and how you fixed it in an attached **REDAME.txt** file.

Tips:

Try to isolate the faulty routine as shown in class by limiting the number of candidate routines in function **create_tc** of the pintool.

Submission requirements:

The submission of this exercise is **in pairs only**.

Submit 1 compressed file called “**ex3.zip**” into the moodle exercise 3 [link](#) containing the following files:

1. The binary of your pintool **ex3.so** (compiled, and tested by you that it runs and gives the result).
2. A directory called: ‘src’ containing all the sources of your pintool along with the make files ‘**makefile**’, **makefile.rules**’ and a **REDAME.txt** file that includes the following:
 - a. names + id numbers
 - b. How to run the tool.

Submission deadline is Sunday, June 21,2026 at midnight.